C2.2 Neural signalling

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-c2-2.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Image credits are on the last slide of the slides for this subtopic.

Neurons at rest

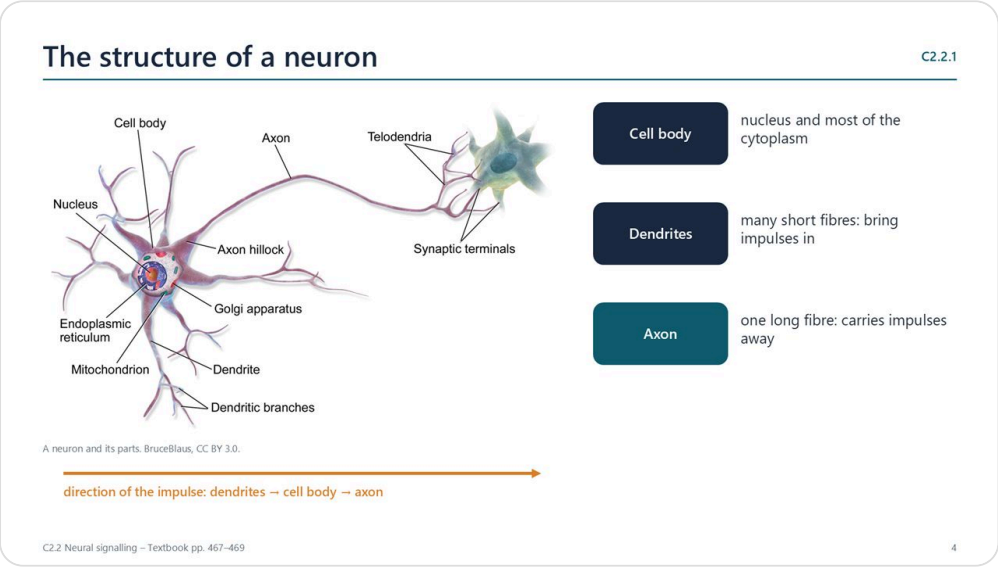
C2.2.1 What neurons are

pp. 467–469

The nervous system is made of cells called **neurons**. The brain and spinal cord form the central nervous system, and peripheral nerves link it with the rest of the body.

A neuron has a **cell body**, containing cytoplasm and a nucleus, with long nerve fibres projecting from it. **Dendrites** are many short fibres that bring impulses to the cell body. An **axon** is one long fibre that carries impulses away. Electrical impulses travel along these fibres.

Sensory neurons bring signals from receptors, relay neurons connect neurons in the CNS, and motor neurons take signals out to effectors, which are muscles and glands.



Watch out: receptors detect stimuli and effectors respond to them.

C2.2.2 How pumping ions sets up the resting potential

pp. 469–470

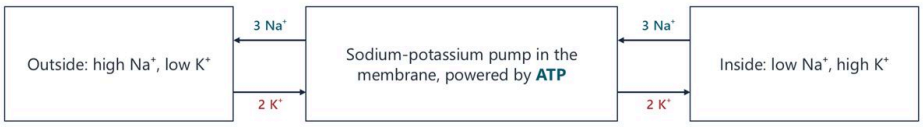
A **membrane potential** is a difference in charge across a membrane. A membrane with positive charge on one side and negative on the other is **polarized**. In a neuron that is not transmitting an impulse, the inside is negative at about **–70 mV**. This is the **resting potential**.

- **The pump:** sodium-potassium pumps use ATP to move three Na^+ out for every two K^+ in. This active transport builds a concentration gradient for each ion, and removes more positive charge than it brings in.
- **Leakage:** the membrane is more permeable to K^+, so K^+ leaks out faster than Na^+ leaks in.
- **Proteins:** large, negatively charged proteins stay inside the axon.

So a "resting" neuron is not idle. It uses ATP all the time to keep the gradients in place.

The sodium-potassium pump

C2.2.2

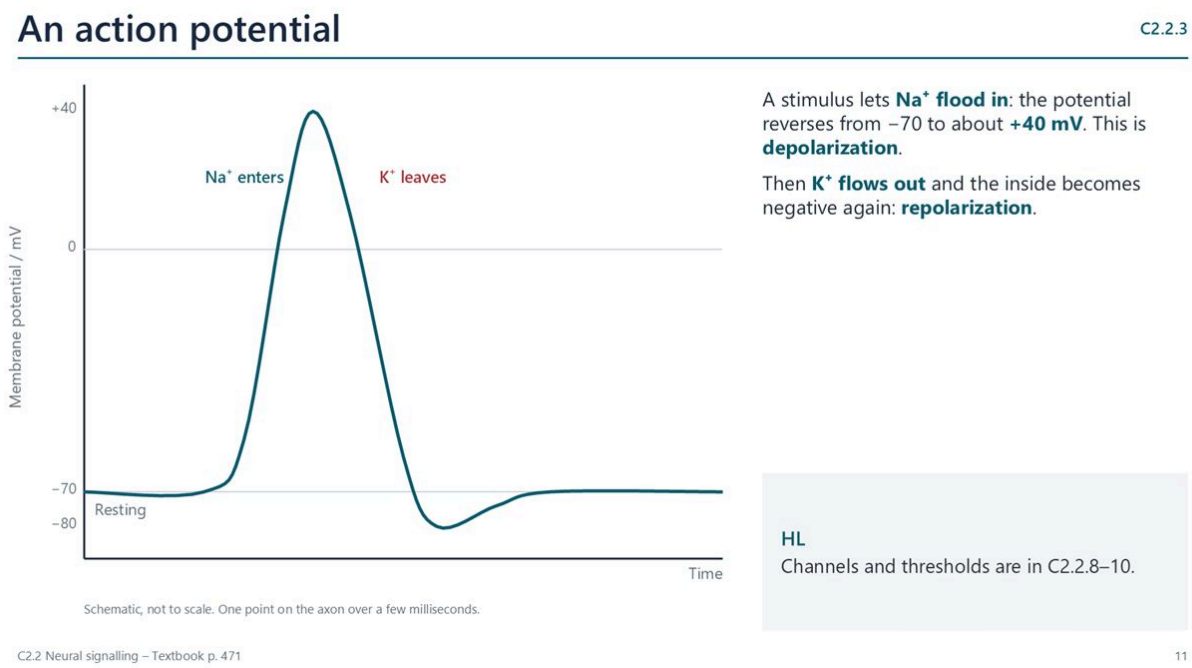


Active transport: 3 Na^+ out and 2 K^+ in for each ATP, against their gradients. This builds **two concentration gradients** and moves more positive charge out than in.

Key idea
ATP comes from aerobic respiration, so nerves need a steady supply of oxygen and glucose even at rest.

The impulse

C2.2.3 Nerve impulses: action potentials travelling along nerve fibres p. 471



A stimulus lets positively charged sodium ions flow into the axon, and the potential reverses from about −70 mV to about +40 mV. Potassium ions then flow out, restoring a negative potential. This brief change is an **action potential**.

An action potential at one point triggers the next region of the fibre, so the impulse is propagated along it. At any one point it lasts only about two milliseconds.

The impulse is **electrical** because it involves the movement of charged ions, which changes the voltage across the membrane. It is not a flow of electrons along a wire.

C2.2.4 Why nerve impulses travel at different speeds pp. 472–475

Fibre	Diameter	Myelin	Typical speed
Small fibre in a mammal	1 μm	No	About 1 m s^{-1}
Squid giant axon	Up to 500 μm	No	About 25 m s^{-1}
Mammal motor fibre	About 20 μm	Yes	About 100 m s^{-1}

Diameter: a narrow fibre has greater resistance to the flow of ions inside it, so a wider fibre conducts faster. Giant axons help a squid to escape quickly.

Myelin: Schwann cells wrap around the axon, forming a lipid-rich sheath with high electrical resistance. Ions cross the membrane only at the gaps, or nodes of Ranvier, so the impulse jumps from node to node. This is far faster, even in much narrower fibres.

Passing the signal on

C2.2.5 Synapses: junctions with neurons and effectors

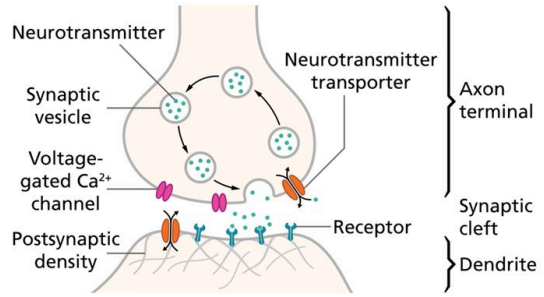
p. 476

A **synapse** is a junction between a neuron and another neuron, or between a neuron and an effector, such as a muscle fibre or a gland. The two cells do not touch. A gap, the **synaptic cleft**, is about 20 nm wide, and an action potential cannot cross it.

Chemical neurotransmitters carry the signal across instead. A signal can cross a typical synapse in only one direction, because vesicles occur only on the presynaptic side and receptors only on the postsynaptic side.

Synapses are junctions

C2.2.5



The **presynaptic** neuron ends in a swollen knob. The **postsynaptic** cell receives the signal.

A **synaptic cleft** about 20 nm wide separates them. An action potential cannot cross it.

A chemical synapse. Thomas Splettstoesser, CC BY-SA 4.0.

Key idea

Chemicals called **neurotransmitters**, stored in vesicles, carry the signal across.

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C2.2.6 How the presynaptic membrane releases neurotransmitter

p. 477

1. An action potential arrives and depolarizes the presynaptic membrane.
2. Calcium channels open, and **Ca^{2+}** diffuses into the knob.
3. Calcium acts as a signalling chemical inside the neuron, making vesicles move to the membrane.
4. Vesicles fuse with the membrane, releasing neurotransmitter into the cleft.

C2.2.7 How an excitatory postsynaptic potential is produced

pp. 477–478

Use **acetylcholine (ACh)** as your example. It is used at many synapses, including the neuromuscular junction.

5. ACh diffuses across the cleft, from high to low concentration.
6. It binds to specific transmembrane receptors on the postsynaptic membrane.
7. These receptors are also ion channels. They open, and Na^+ flows in.
8. This depolarizes the membrane: an **excitatory postsynaptic potential**. If it is large enough, it starts an action potential in the next cell.
9. The enzyme acetylcholinesterase breaks ACh down to choline and acetate. The channel closes, and the choline is reabsorbed and reused.

Watch out: always say that the neurotransmitter crosses the cleft by **diffusion**. Remember to mention its removal by an enzyme.

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